

Integration of diagnosis and treatment of sleeping sickness in primary healthcare facilities in the Democratic Republic of the Congo

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Abstract

BACKGROUND Control of human African trypanosomiasis (HAT) in the Democratic Republic of Congo (DRC) has always been a vertical programme, although attempts at integration in general health services were made in recent years. Now that HAT prevalence is declining, the integration question becomes even more crucial. We studied the level of attainment of integration of HAT case detection and management in primary care centres in two high-prevalence districts in the province of Bandundu, DRC.

METHODS We visited all 43 first-line health centres of Mushie and Kwamouth districts, conducted structured interviews and inspected facilities using a standardised checklist. We focused on: availability of well trained staff – besides HAT, we also tested for knowledge on tuberculosis; availability of equipment, consumables and supplies; and utilisation of the services.

RESULTS All health centres were operating but most were poorly equipped, and attendance rates were very low. We observed a median of 14 outpatient consultations per facility (IQR 8–21) in the week prior to our visit, that is two patients per day. The staff had good knowledge on presenting symptoms, diagnosis and treatment of both HAT and tuberculosis. Nine centres were accredited by the national programme as HAT diagnosis and treatment centres, but the most sensitive diagnostic confirmation test, the mini-anion exchange centrifugation technique (mAECT), was not present in any. Although all nine were performing the CATT screening test, only two had the required cold chain in working order.

CONCLUSION In these high-prevalence districts in DRC, staff is well-acquainted with HAT but lack the tools required for an adequate diagnostic procedure. Attendance rates of these primary care centres are extremely low, making timely recognition of a resurgence of HAT unlikely in the current state of affairs.

keywords human African trypanosomiasis, Democratic Republic of Congo, primary healthcare facilities

Introduction

Human African trypanosomiasis (HAT) is a vector-borne parasitic disease caused by two subspecies, *Trypanosoma brucei* (*T.b.*) *gambiense* and *T.b. rhodesiense*, both transmitted by the tsetse fly (Brun *et al.* 2010). HAT is usually fatal if untreated. Ninety-eight per cent of HAT cases are caused by *T.b. gambiense*, which occurs in 24 countries, all in sub-Saharan Africa (WHO 2014), and it is

estimated that 57 million people live in areas at risk for *T.b. gambiense* HAT. The current annual HAT incidence is estimated at below 20 000 (Franco *et al.* 2014; Simarro *et al.* 2011, 2012). In 2012, 7216 cases were notified, 83% of which were from the Democratic Republic of Congo (DRC) (WHO 2014). In recent years, there has been a sharp decline in case notification rates of HAT in all endemic countries. WHO has now scheduled HAT for elimination as a public health problem by 2020. The

main strategy is based on early case detection and treatment, case detection is either active or passive. Active case detection by mobile screening teams is to be conducted in all historic HAT foci until no more cases are detected at village level over a 5-year period. From that time onwards, screening should be shifted to passive, making use of existing general healthcare services. Simarro *et al.* (2014) assessed the healthcare infrastructure in HAT-endemic areas and estimate that more than 80% of the population at risk for *T.b. gambiense* HAT live within 5-h travel of a fixed health facility offering diagnosis and treatment for HAT. Over the period of 2000–2012, approximately 50% of all notified HAT cases were diagnosed passively (Simarro *et al.* 2014).

The debate on the merits of integrated (horizontal) *vs.* vertical approaches of disease control has been going on since the 1960s and so far no consensus has been reached. Dudley and Garner (2011) believe that the two approaches are not opposed, but complementary to each other. Vertical programme services can be integrated into primary health care, which implies that multipurpose health services take responsibility for implementing control activities related to a specific disease or health problem. Integration can be operational, administrative or complete (both administrative and operational), depending on which responsibilities are handed over to the general healthcare system (Debrouwere & Pangu 1989). The expected benefits of the integrated approach are gains in efficiency, improvement of health services quality, greater coverage of the population and reduced disparities in access and utilisation of services (Dudley & Garner 2011; Shigayeva *et al.* 2010). In the literature, a number of prerequisites for successful integration have been listed: (i) the health services must be sufficiently functional; (ii) polyvalence of health staff must be guaranteed; (iii) integration should not be attempted when disease prevalence is low because health staff would not see enough patients to maintain their specific technical competence for dealing with the problem and will be reluctant to undergo specific training for a rarely occurring problem; (iv) decision-making power should be transferred to the polyvalent health services while respecting the technical standards issued by the programmes; and (v) the benefits should be mutual: better coverage for programmes and better response to patient demand for the patient-centred primary care services (Criel *et al.* 1997, 2004; Dudley & Garner 2011; Pepin *et al.* 1989).

Case finding and treatment of HAT in the DRC has traditionally been organised as a fully vertical programme. Case detection is performed by mobile screening teams moving from village to village and screening entire populations (WHO 2013). Previously, HAT cases

identified were referred to fixed specialised health facilities providing HAT treatment only; in recent years, HAT treatment has been gradually integrated into general health facilities and an increasing number of general health facilities now also provide HAT diagnostic services. The first attempts to integrate management of HAT into the general health services date from the 1980s (Laveissiere & Meda 1993; Mercenier 1977; Pepin *et al.* 1989). However, the complexity of diagnostic procedures, requiring parasitological confirmation by microscopy and a lumbar puncture for stage determination, limits feasibility (WHO 1998). Even the CATT screening test is difficult to apply under routine health centre conditions because it requires refrigeration and an electric rotor. Moreover, it is not an individual test as it is manufactured in vials of 50 tests that quickly expire once opened (Mitashi *et al.* 2012). In the recent HAT elimination strategy of WHO, its panel of experts promotes integrating surveillance activities into general health services by recommending the use of rapid tests in health centres. But so far none of the health facilities in DRC is equipped with these new tools; thus, CATT remains the sole screening test in use. The current treatment of HAT consists of daily intramuscular injections of pentamidine for stage 1 or twice-daily IV infusions of eflornithine combined with oral administration of nifurtimox (NECT) for stage 2. NECT is far less toxic (Alirol *et al.* 2013; Schmid *et al.* 2012) than the treatment regimen used earlier for stage 2 HAT, but is still too complex to be administered in the daily routine of most peripheral health facilities (Yun *et al.* 2010).

In recent years, there has been substantial investment in diagnostic tools and treatment for HAT. NECT has now become the standard first-line treatment for stage 2. A fully oral treatment regimen for both stages of the disease, obviating the need for a lumbar puncture, is in an advanced stage of clinical evaluation (Tarral *et al.* 2014). Two individual point-of-care screening tests, which can be stored at ambient temperature, are user-friendly, require minimal training (Buscher *et al.* 2013; WHO 2013) and have also become available. These developments, and a sharp reduction in the numbers of HAT cases reported annually (WHO 2013), have led to renewed interest in integration of HAT diagnosis and treatment into general healthcare services. The question remains whether the prerequisites for successful integration are being met, in particular the need for an adequate level of performance of the general health services and the fact that the prevalence of the disease should not be too low (Criel *et al.* 1997).

Bandundu province in the DRC currently reports the largest numbers of HAT cases of all endemic countries.

In two of the most HAT-endemic districts of this province, Mushie and Kwamouth, efforts towards integration of diagnosis and treatment of HAT into general healthcare services have started some years ago. The aim of our study was to see how effective this integration has been and which obstacles have been encountered. As a comparison, we included tuberculosis diagnostic and treatment practices in our assessment as the tuberculosis programme has a much longer history of integrated services.

Material and methods

The study was designed as a health centre survey and conducted in Kwamouth and Mushie, two highly HAT-endemic districts in the province of Bandundu in the DRC (Figure 1). Both are rural districts with populations estimated at 116 434 for Kwamouth and 115 048 for Mushie. The health services are organised along the lines of a district health system with one hospital per zone supported by a number of health centres: Kwamouth has 22 health centres, Mushie has 21. Of those, 3 health centres in Kwamouth and 2 in Mushie provide tuberculosis diagnostic services.

Specific HAT diagnostic services are provided by 6 centres in Kwamouth and 3 in Mushie (Table 1), and henceforward, these nine centres are called 'designated HAT centres'. They were identified by the national HAT control programme, staff was trained, and equipment and supplies were provided. Stage 2 HAT treatment is provided by only four centres in Kwamouth and 1 in

Table 1 Number of health centres providing diagnosis and treatment of HAT in both health districts

Items	Kwamouth (<i>n</i> = 22)	Mushie (<i>n</i> = 21)
1. HAT		
Clinical diagnosis	22 (100%)	21 (100%)
CATT screening*	6 (27%)	3 (14%)
Parasitology classical methods†	6 (27%)	3 (14%)
Parasitology CTC or mAECT	0 (0%)	0 (0%)
Treatment Stage 1‡	22 (100%)	21 (100%)
Treatment Stage 2‡ (NECT)	4 (19%)	1 (5%)
2. TB		
Clinical diagnosis	22 (100%)	21 (100%)
Ziehl–Neelsen§ microscopy	3 (14%)	2 (10%)
First-line treatment‡	22 (100%)	21 (100%)
Second-line treatment‡	0 (0%)	0 (0%)

*Of all those, only two have a functional cold chain.

†Of those, six have a microscope in good working order.

‡Supplied on a needs basis; in both districts, treatment is also available at district hospital.

§Of those, three have a microscope in good working order.

Mushie; stage 1 HAT treatment can in principle be provided in any health centre in these endemic districts. The health centres are typically run by one or two nurses. While some centres are accessible by road, most need to be accessed from the rivers.

In June and July 2013, we visited each health centre and conducted a structured interview with the nurse in charge. The assessment framework was adapted from Kyabayinze *et al.* (2012). We assessed availability and

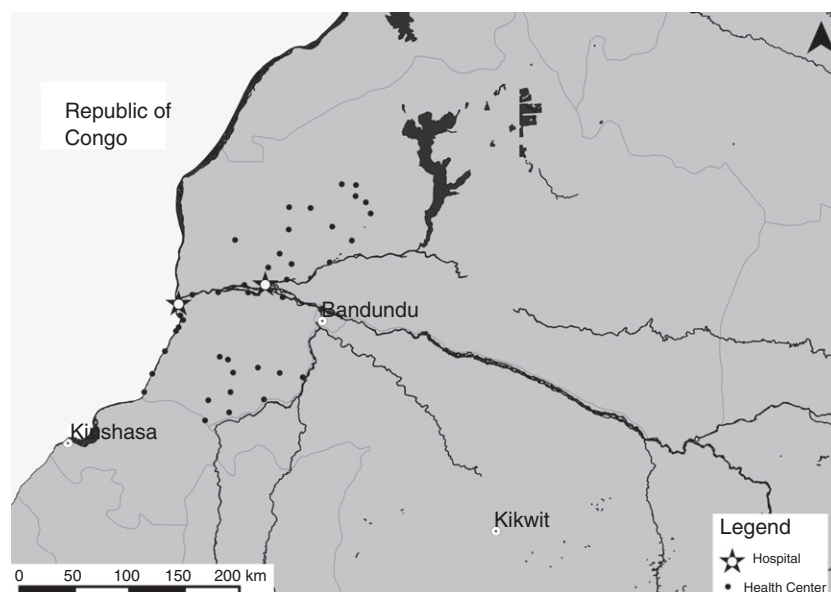


Figure 1 Map of Bandundu, highlighting the study area (Kwamouth and Mushie health districts).

knowledge of health services staff; availability of equipment, consumables and supplies; and utilisation of the services.

Availability of staff was assessed by their presence in the centres upon our arrival, which was during working hours but unannounced. Knowledge was assessed through a standardised questionnaire which included questions on mode of transmission, presenting symptoms, diagnostic procedures and treatment of both HAT and tuberculosis. All questions were open-ended and interviewees were not provided with pre-formulated answers to choose from.

The availability of equipment, consumables and supplies was assessed by completing a checklist which included essential diagnostic equipment (e.g. microscopes, centrifuges, CATT rotators) and consumables (e.g. Ziehl-Neelsen staining reagents, Giemsa stain, mAECT columns). Availability of treatment for HAT and tuberculosis was assessed by inspecting facility drug stores. We also inspected the buildings and recorded whether or not they had electricity and a functional cold chain. A microscope was considered functional if a thick film slide brought along by the investigator could be read using the oil immersion objective.

To assess the utilisation of the services, we recorded the numbers of outpatient consultations in the week prior to our visit from the outpatient register. We also obtained annual outpatient attendance figures and catchment area population sizes from the provincial coordination to triangulate the results obtained from the facility registers. As part of the interviews, staff were also asked about the numbers of cases of suspected HAT and tuberculosis, referred and/or diagnosed and treated in the 3 months preceding the visit. We also inspected various cards and registers, that is outpatient register, laboratory register, specific laboratory registers for HAT and tuberculosis, and treatment registers and treatment cards for HAT and tuberculosis. We noted whether these forms and registers were present and when the last entry had been made.

Data were double-entered in an EPI-info database and analysed using STATA version 11 (StataCorp LP, College Station, TX, USA).

Ethics approval

The protocol of this study was formally reviewed and approved by the Institutional Review Board of the Institute of Tropical Medicine (ITM) and by the Ethics Committees of the University Hospital in Antwerp and of the School of Public Health at the University of Kinshasa.

Results

We visited all 43 health centres of two HAT-endemic districts and interviewed all 43 staff members in charge. Forty centres were run by nurses, three by doctors. In most centres, staff were present upon our arrival, and in seven centres (16%), we had to wait for several hours for the staff to arrive. The centres were typically housed in mud-walled buildings, with roofs of straw or corrugated metal sheets; four centres were on the verge of collapse. Furnishing was minimal; none of the centres had beds. Seven centres (16%) had electricity, 10 centres (23%) had a functional cold chain. Outpatient attendance rates were generally low (median number of 14 consultations in the week preceding our visit (i.e. 2 per day), interquartile range [IQR] 8–21). Extrapolating the observed consultation rate to an annual rate results in a utilisation rate of 0.16 per inhabitant. From the records provided by the provincial health authorities, the average annual number of visits to any of the centres varied from 165 to 4602. The reported utilisation rate for 2012 was 0.26 per inhabitant per year for the two districts combined.

Functional microscopes with oil immersion objective were present in 25 centres (58%) (Table 2), and these included 6 of 9 (67%) designated HAT diagnostic centres and 3 of 5 (60%) centres designated for tuberculosis

Table 2 Presence of HAT and TB related equipment, consumables and drugs in designated and non-designated facilities

Items	Health centres designated for HAT diagnosis (<i>n</i> = 9)	Non-designated health centres (<i>n</i> = 34)	Health centres designated for TB diagnosis (<i>n</i> = 5)
1. Equipment			
Microscope*	6 (67%)	16 (47%)	3 (60%)
CATT rotator	6 (67%)	0 (0%)	NA
Refrigerator	3 (33%)	1 (3%)	NA
Centrifuges	1 (11%)	0 (0%)	NA
Microscopes slides	9 (100%)	33 (97%)	5 (100%)
Slides boxes	4 (44%)	13 (38%)	5 (100%)
2. Consumables			
Immersion oil	8 (89%)	11 (32%)	5 (100%)
Giemsa	8 (89%)	12 (35%)	NA
Distilled water	0 (0%)	1 (3%)	0%
Ziehl-Neelsen kit	NA	NA	5 (100%)
3. Drugs			
Pentamidine	6 (67%)	1 (3%)	NA
NECT	3 (33%)	0 (0%)	NA
First-line drugs			5 (100%)

*In good working order.

diagnosis. Rotators required for the CATT screening test were present in 6 of 9 centres designated for the HAT diagnosis (67%), of those 3 (50%) had a functional cold chain. Only 1 centre had an electric centrifuge, required for the most sensitive HAT diagnostic test, the mAECT. The mAECT test itself was not available in any of the centres.

Reagents required for Ziehl–Neelsen staining were available in all centres involved in tuberculosis diagnosis, although one lacked sulphuric acid. Giemsa, required for thick films for malaria but also for HAT diagnosis, was present in 20 health centres (47%) (Table 2), of which 16 also had a microscope with functional oil immersion objective. These included 6 (67%) of the 9 designated HAT diagnostic centres.

At the time of the visits, treatment for stage 1 of HAT was available in 7 centres (16%), all of which were designated HAT treatment centres (Table 2). In all other centres, it is made available on an ad hoc basis. Stage 2 treatment (NECT) was available in 3 of 9 designated HAT centres (33%).

Of all centres visited, 26 (60%) had referred at least one suspect case of tuberculosis in the 3 months preceding the visit; for HAT, this was the case in 17 centres (40%).

Of the 5 centres involved in tuberculosis diagnosis, three (60%) had diagnosed a case of tuberculosis during this period, whereas 7 of 9 designated HAT centres (78%) had diagnosed at least 1 case of HAT. Twenty-one centres (49%) had treated a case of tuberculosis in the past 3 months, while 8 (19%) had treated a case of HAT (Table 3).

Knowledge on tuberculosis and HAT was generally good among staff interviewed. The mode of transmission was well known for both diseases. For HAT, all responded correctly that transmission is vectorborne; for tuberculosis 41 (95%) correctly replied that transmission is airborne, although 12 (28%) also thought that the disease can be spread through direct contact.

All interviewees mentioned cough as main presenting symptom for tuberculosis, followed by weight loss (42, i.e. 98%) and fever (39, i.e. 91%) (Table 4). For HAT, there was more diversity in the responses, but symptoms mentioned were by and large correct. Most often

Table 4 Presenting signs of tuberculosis and HAT mentioned by health centre staff ($n = 43$)

Disease			
Tuberculosis		HAT	
Signs	Number (%)	Signs	Number (%)
Cough	43 (100)	Headache	38 (88)
Weight loss	42 (98)	Abnormal behaviour	38 (88)
Fever (night sweats)	39 (91)	Somnolence	34 (79)
Loss of appetite	37 (86)	Weight loss	32 (74)
Dyspnoea	11 (26)	Fever	30 (70)
Haemoptysis	10 (23)	Lymphadenopathy	20 (47)
Chest pain	7 (16)	Itching	19 (44)
		Amenorrhoea	9 (21)
		Convulsions	3 (7)
		Tremors	3 (7)

mentioned were headache (88%), abnormal behaviour (88%), somnolence (79%) and fever (70%) (Table 4).

When asked about diagnosis of the two diseases, smear microscopy for tuberculosis was mentioned by 41 (95%) and 10 (23%) mentioned X-ray. For HAT, the diagnostic test most often mentioned was CATT (98%), followed by lymph node puncture (81%), thick film (79%) and fresh blood examination (74%). All these tests were mentioned by all staff from designated HAT diagnostic centres. Twenty-two (61%) mentioned lumbar puncture as diagnostic test; mAECT was mentioned by 5 (12%), of whom 4 were from designated HAT centres.

Knowledge on treatment for tuberculosis was very good; 98% mentioned the correct treatment duration and although treatment is provided as a fixed drug combination, most knew at least two of the four generic names of the drugs (rifampicin 37, i.e. 86% and isoniazid 36, i.e. 84%). For HAT, all staff were aware of the fact that treatment regimens vary according to disease stage and 37 (86%) spontaneously mentioned that stage determination is performed by lumbar puncture. Twenty (47%) correctly mentioned the current treatment regimen for stage 1, eight (19%) knew the correct treatment for stage 2, and these included four of the designated HAT treatment centres (80%). The exact dosing schemes and duration of treatment were mentioned by 20 (47%) for stage 1 *vs.* eight (19%) for stage 2.

Discussion

We visited 43 health centres that were all operating but under-utilised and mostly in a poor state of maintenance.

Table 3 Involvement in HAT and tuberculosis management in health centres visited ($n = 43$)

Events in past 3 months	Tuberculosis (%)	HAT (%)
Referred at least one suspect case	26 (61)	17 (40)
Diagnosed at least one case	17 (40)	7 (16)
Treated at least one case	5 (12)	8 (19)

Median outpatient attendance in the week preceding our visits was only two patients per day. Our visit was unannounced, yet in 84% of the centres, staff were present, in the remainder they arrived with some delay. Knowledge and awareness about both tuberculosis and HAT among the staff were generally very good. Although most centres designated for HAT and tuberculosis diagnosis do have a functional microscope, there is still a serious lack of essential equipment.

The fact that 2 of 5 designated tuberculosis diagnostic centres had not diagnosed a single case in the 3 months preceding our visits raises some concerns over their ability to maintain their technical skills in smear microscopy. HAT cases had been diagnosed by 7 of 9 designated centres over the preceding 3 months. This is encouraging, although one may wonder about how many cases have been missed due to suboptimal conditions. Most of the designated HAT centres offering the CATT test do not have a functional cold chain, which in all likelihood will compromise the quality of antigen and therefore the validity of test results. Centrifuges required for the more sensitive parasitological confirmation tests were present in only 1 of 9 designated HAT centres and absent in all other centres. Microscopy is usually performed without a source of electricity, which reduces the chance of seeing either parasites or acid-fast bacilli.

Considering the five conditions for successful integration, there appear to be problems with functional health services and a prevalence that is not too low. With a median outpatient attendance of two per day, one may question whether the system is really functional; certainly such low attendance rates will interfere with the capability of the system to detect a HAT resurgence at an early stage (WHO 2013).

Another issue related to the functionality of the system is the generally poor provision of equipment and supplies. All HAT patients identified were diagnosed with classical methods such as lymph node puncture and fresh blood film because capillary centrifugation technique (CTC) and mAECT were not available. Sensitivity of these classical methods is low, generally below 50% (Lutumba *et al.* 2006) and likely to be even lower if performed by staff who do not see trypanosomes regularly and who use mediocre equipment. In theory, 22 of the 43 health centres could provide HAT diagnosis because they do have functional microscopes. In reality, only six would be able to screen patients systematically because none of the others has the CATT screening test available. Of these six, all of whom are designated HAT diagnostic and treatment centres, only three have a functional cold chain, which is a prerequisite for use of the CATT test. Taking these constraints into account, the proportion of the

population at risk living within 5-h travel of a fixed health facility offering diagnosis and treatment for *T.b. gambiense* HAT is probably much below the 80% reported by Simarro *et al.* (2014).

If HAT diagnostic services are decentralised, there certainly is a need for a reliable quality assurance system. In tuberculosis control, such systems are already widely practiced, usually based on blinded rereading of microscopy slides (Ridderhof *et al.* 2007). A recent external quality assessment of microscopy carried out in DRC showed that only 49% of laboratories could recognise trypanosomes and only 20% could perform adequate Giemsa staining (Mukadi *et al.* 2011). It is therefore very well conceivable that a HAT patient presenting to such facilities will incur substantial health systems delay or not be diagnosed at all (Hasker *et al.* 2011).

Staff not seeing trypanosomes on a regular basis is related to the third condition for successful integration: 'a prevalence that is not too low'. The disease burden should be sufficient to maintain an acceptable level of diagnostic and/or therapeutic expertise. Even in these highly endemic districts, HAT prevalence is below 0.5%. It may be higher among those consulting the health services, but if the average centre sees fewer than 800 outpatients per year, the number of HAT patients among them is bound to be small. In less endemic districts, prevalences are generally below 0.1%, lowering the probability of a health worker being confronted with a HAT patient even further.

The newly developed rapid diagnostic tests (RDTs) could play an important role as these are individual thermostable tests that are easy to perform (Buscher *et al.* 2013). Although these tests seem attractive for screening, they must be assessed in large-scale effectiveness studies before their introduction in health centres. A system in which health centre staff identify suspect cases which are then referred to an expert for confirmation seems the most feasible option. The newly developed LAMP performed on a filter paper sample could be considered as a second step of screening to avoid unnecessary referrals for confirmation, but it needs to be validated first under conditions prevailing at district level. Convertible LED microscopes and centrifuges operating on photovoltaic power are also available at reasonable cost. Introducing such equipment would be beneficial both for HAT and TB diagnosis. Another option would be a visiting supervisor, as is practiced in leprosy control programmes (Roos *et al.* 1995).

A limitation for our study is that we assessed only the health centres and not the district hospitals. In all probability, substantial numbers of patients will bypass the health centres and report directly to the district hospitals. Moreover, utilisation rates obtained by extrapolation

from figures for the week preceding our visits may be an underestimate. Consultation rates may differ depending on the season, which could explain the higher utilisation rates reported by the provincial coordination. Our study did not assess the supervision system, but for any integrated approach to work the health system will need to be strengthened. Even if, as shown by Simarro *et al.* (2014), 80% of the population at risk live within 5-h travel of a health facility able to diagnose HAT, this does not mean that patients will present for consultation or that those who do so will be adequately diagnosed. For the time being, a need for active case finding remains.

Conclusion

Integration of HAT case finding and treatment is possible but requires investments in the primary healthcare services. Given the complexity of HAT diagnosis and the low prevalence of HAT, even in this highly endemic area, a 2-tier system of identifying suspects serologically by a RDT followed by diagnostic confirmation and treatment at a higher level seems most appropriate. If a safe oral treatment becomes available and RDTs are shown to perform well in the daily routine of health centres, HAT patients could be treated at health centre level on the basis of a positive RDT. Lack of specificity of the tests would then be offset by the use of safe and effective oral drugs. For epidemiological surveillance of historic foci, an active approach seems mandatory until healthcare services have been sufficiently strengthened.

Acknowledgements

We gratefully acknowledge the Professor Guy Kegels for proofreading the manuscript.

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